



Semester 1 Examinations 2014/2015

Exam Code(s)
Exam(s) Higher Diploma in Software Design and Development

Module Code(s) CT537

Module(s) Software Engineering Methods

Paper No. 1
Repeat Paper

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Instructions: Candidates should answer **Question 1** and any **two other** questions.
All questions carry equal marks.

Duration 2 hours

No. of Pages 4

Requirements:

MCQ
Handout
Statistical/ Log Tables
Cambridge Tables
Graph Paper
Log Graph Paper
Other Materials

Release to Library: Yes ☐ No ☐

1. Draw a **context diagram**, a **systems level DFD** and one **third level DFD** to represent the data and processing involved in the following system description for a travel company. *Explorer Tours* specialises in offering tailor-made exploration holidays. A website is being developed that will allow customers to pre-book a holiday online for a specific period. The website will provide the following functionality at a minimum:

- Accept customers' personal details, select preferred destinations and dates and provide details of the number of persons travelling in the group.
- Accept any discount code customers may have and details of previous travel with *Explorer Tours* before (Note that returning customers are to be offered a 5% discount on holiday packages).
- Display a series of holiday packages ordered by increasing price (includes flight and hotel details).
- The user can select the most preferred package and request any further information they require.
- Accept customer payment: either a deposit or full electronic payment, and email an order confirmation number.
- Ensure all holidays are paid for in full at least two weeks before the departure date.
- Handle any cancellations or complaints from customers.

(20)

2. (a) During system design, different criteria are used to evaluate the design of the presentation (UI), the application (business logic) and the data layers. Compare the different evaluation requirements for each of these system layers, using examples to illustrate your answer.

(10)

(b) In structured analysis and design, what is the relationship between the ERD and DFD models? Illustrate your answer with examples from the *Explorer Tours* system explored in Q.1.

(5)

(c) What is scope creep? How should it be addressed to ensure project success?

(5)

3. (a) Compare and contrast the Structured Analysis DFD and the UML's Activity Diagram as two examples of process modelling techniques.

(5)

(b) You are the team lead of a systems design team that is implementing a multiple module project using new technologies with an expected duration of 2 years. Your client just sent you the memo included below. Write a memo back to your client answering their questions.

Dear Lead,

I was reading **IT Update** magazine and an article mentioned that many project teams now use prototyping and agile techniques to speed up their systems development processes. What are these techniques? Will they save us time and money? What problems might we encounter if we use them?

Regards,
TJ Murphy.

(9)

(c) Provide a brief report detailing how the **implementation** of the Travel Agency system in Q.1 will be approached. The report should detail the design method, programming language and framework employed, as well as the implementation strategy stating all assumptions made.

(6)

4. (a) In his book *Controlling Software Projects*, Tom DeMarco states that “you cannot control what you cannot measure.”

(i)What does this statement mean for software project managers?

(ii)Your manager assigns you to plan the collection of software metrics in your software development company. What questions might you ask before you suggest any specific metrics to collect? What metrics would you collect?

(8)

(b) How would you define software quality? Evaluate the contribution of software engineering techniques to improving the quality of software produced

(4)

(c) Prepare a **requirements search strategy** for a new patient management information system you are developing for a busy GP practice. Clearly indicate the stakeholders and the requirements gathering techniques you have chosen, ensuring you have considered identification of both functional and non-functional requirements in your outlined strategy.

(8)